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Department of Information Technology

A.Y. 2025-2026
Class: BE-IT A/B, Semester: VIII
Subject: Cloud Computing Lab

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Experiment – 4: Software as a Service (SaaS)

Aim: To study and Implement Software as a Service using AWS Rekognition to detect, analyze and compare images and videos. (Free Tier Cloud Platforms).

Objective: After experimenting, the students will be able to –
- to upload images and videos to detect them for emotional identification,
- to compare images with existing images and measure the similarity score.

Lab objective mapped : ITDO8024.2: To learn, simulate, and examine cloud service model and deploy IAAS, SAAS, and PAAS services over different cloud platforms.

Prerequisite:
An AWS account with an active subscription.
Install Visual Studio Code (For API calling and programming)

Requirements: Cloud Login**,** Desktop, Browser, Internet etc.

Pre-Experiment Theory:

What is Amazon Rekognition?

Amazon Rekognition is a service that makes it easy to add powerful visual analysis to your applications. Rekognition Image lets you easily build powerful applications to search, verify, and organize millions of images. Rekognition Video lets you extract motion-based context from stored or live stream videos and helps you analyze them.

Rekognition Image is an image recognition service that detects objects, scenes, activities, landmarks, faces, dominant colors, and image quality. Rekognition Image also extracts text, recognizes celebrities, and identifies inappropriate content in images. It also allows you to search and compare faces.
Rekognition Video is a video recognition service that detects activities, understands the movement of people in frame, and recognizes objects, celebrities, and inappropriate content in videos stored in Amazon S3 and live video streams. Rekognition Video detects persons and tracks them through the video even when their faces are not visible, or as the whole person might go in and out of the scene. For example, this could be used in an application

that sends a real-time notification when someone delivers a package to your door. Rekognition Video allows you also to index metadata like objects, activities, scene, landmarks, celebrities, and faces that make video search easy.

Procedure

1. Login to AWS console
2. Move on to AWS Rekognition service console
3. Choose facial analysis feature from the left navigation panel. It allows analyzing faces in an image and receiving a JSON response.
4. Use any sample group image of your choice
5. Click the orange upload button
6. Select sample image
7. Observe result using result dropdown
8. See results for different faces
9. Use response dropdown to see JSON results.
10. Notice emotions results - It shows confidence ratings in percentage form
11. Select face comparison in the panel navigation in the left panel
12. Use single sample image
13. Click on upload option for reference face and select sample image
14. Click on orange upload button for comparison face, select another sample image
15. Notice results
16. Check response section
17. Similarity score- ranges from 1-100, threshold can be adjusted when using Rekognition API

Post-Experiments Exercise

Extended Theory:

1. Image formats that Amazon Rekognition support?(select correct option)

- ☐ JPEG
- ☐ PNG
- ☑ both JPEG AND PNG

2. The H.264 codec is commonly used for the recording, compression and distribution of video content.

3. The video must be encoded using the H.264 codec video format for Amazon Rekognition? (select correct option)

- ☑ TRUE
- ☐ FALSE

Results/Calculations/Observations:
Fill the following observation tables-(to be written in hand)

Sr. No	Questions	Do as directed
1.	What are the most common use cases for Amazon Rekognition? Choose all correct ans	☐ Searchable Image Library ☐ Face-Based User Verification ☐ Sentiment Analysis ☐ Facial Recognition ☐ Image Moderation
2.	The most common use cases for Rekognition Video include 1)Search Index for video archives and 2) Easy filtering of video for explicit and suggestive content	☐ TRUE ☐ FALSE

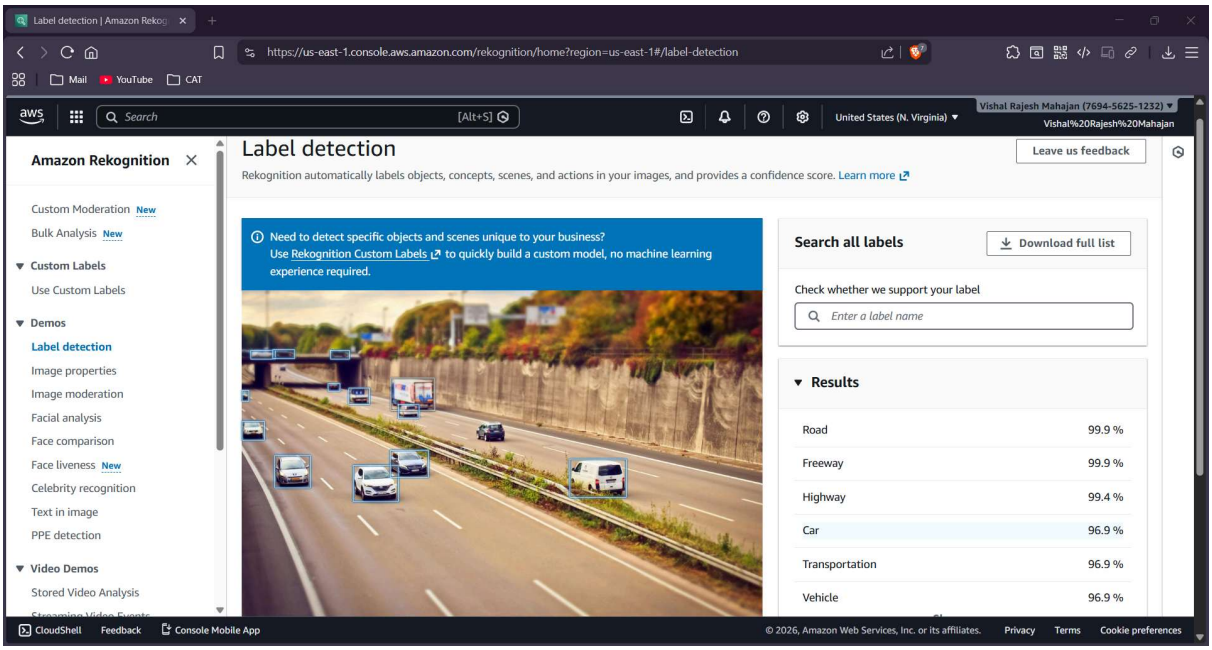
Post Experimental Exercise-

Conclusion:
1. Write what was performed in the experiment
2. Mention a few applications of what was studied.
3. Write the significance of the studied topic

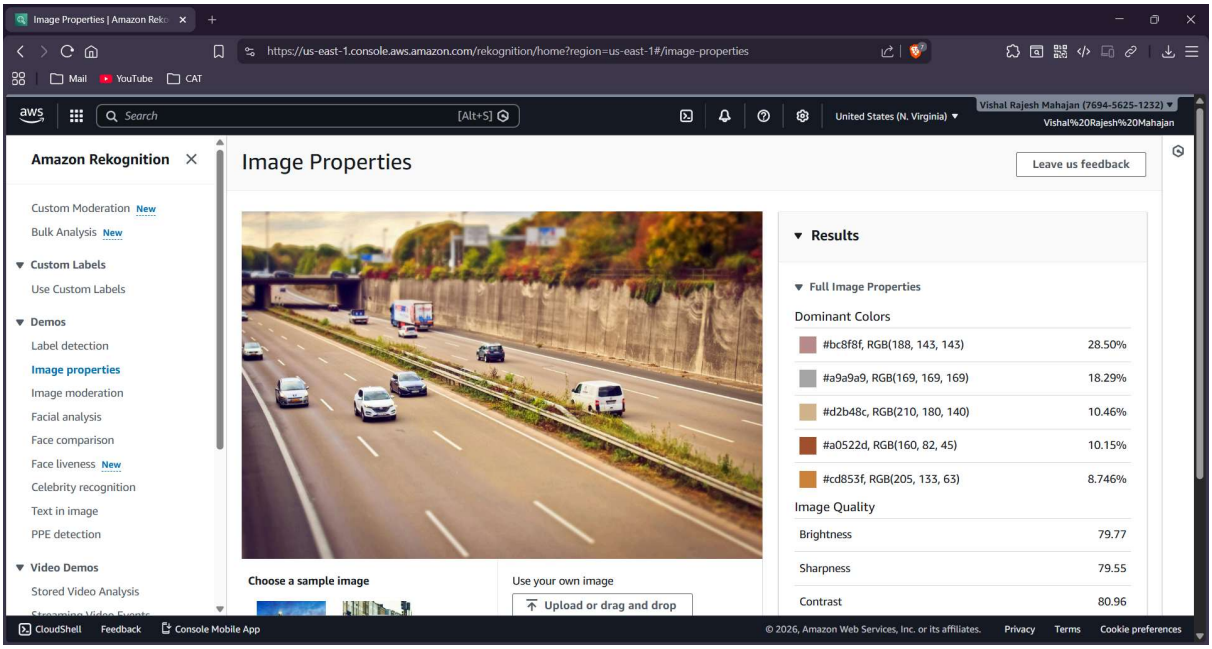
References:
[1] [Online]https://learn.microsoft.com/en-us/azure/app-service
[2] [Online]https://learn.microsoft.com/en-us/azure/app-service/quickstart-nodejs
[3] [Online] https://stackify.com/top-azure-paas-services-developers/

Pre Experiment Exercise:

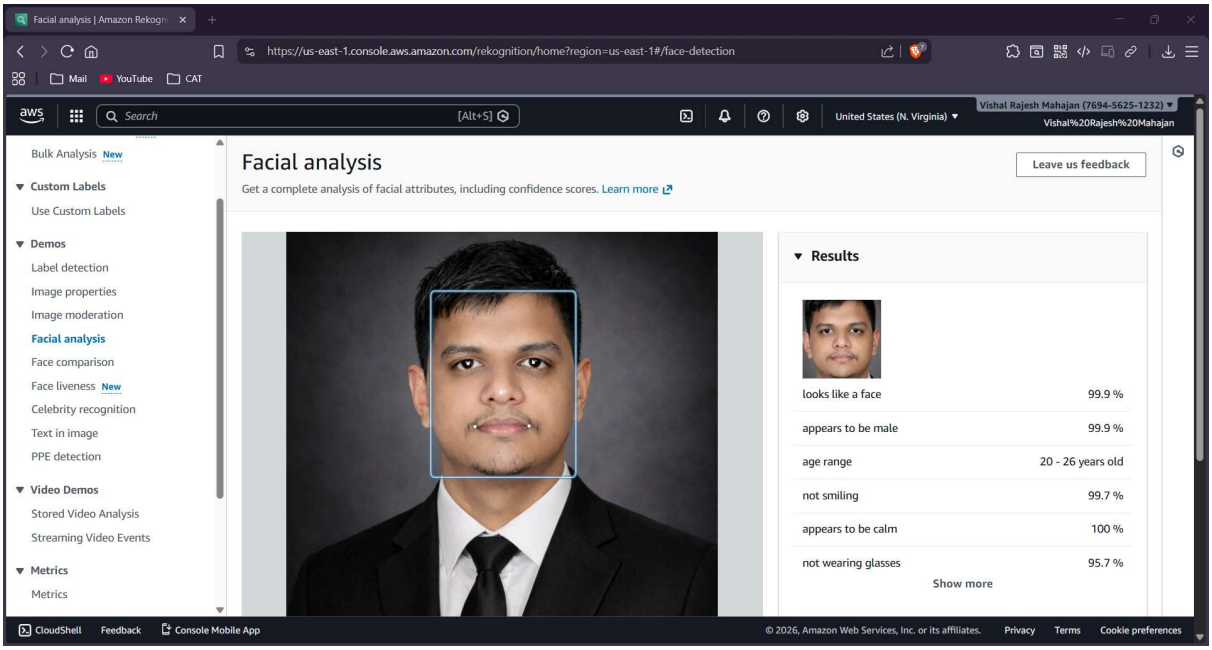
1. Label Detection



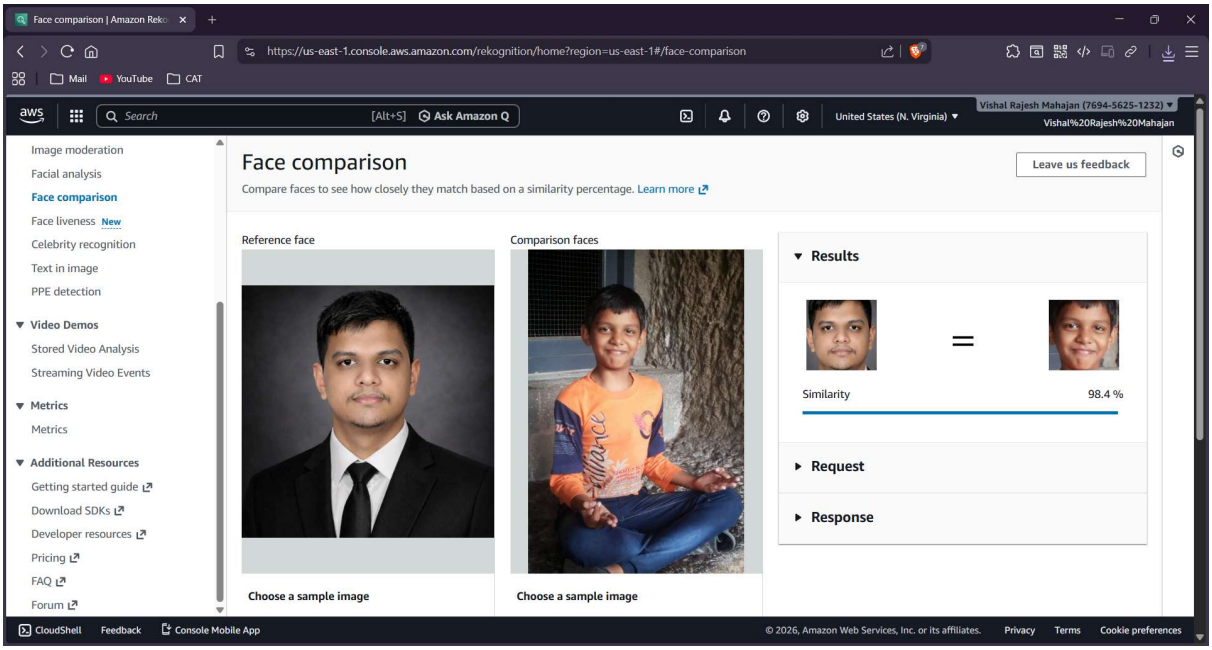
2. Image Properties



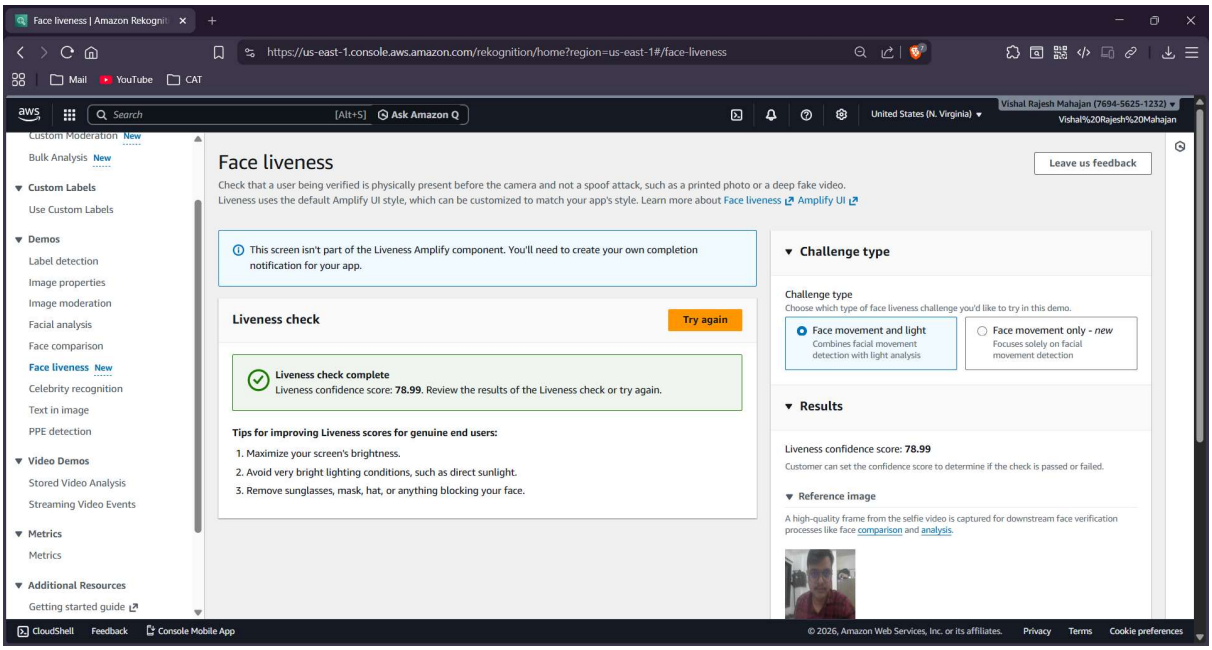
3. Facial Analysis



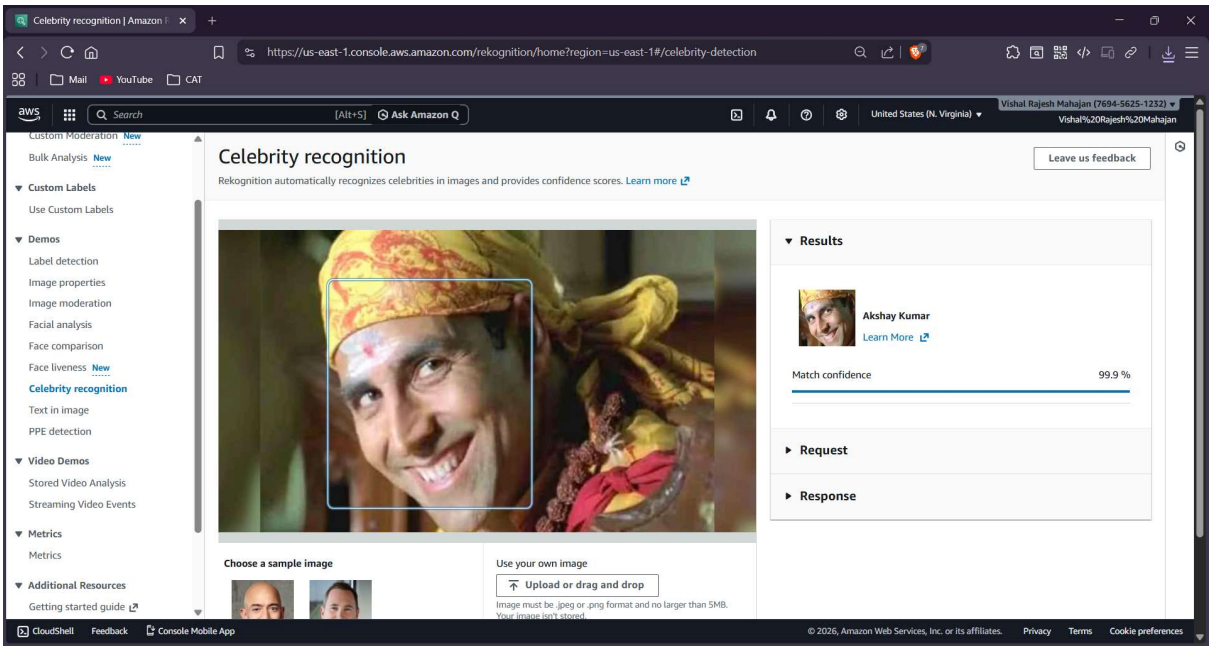
4. Face Similarity



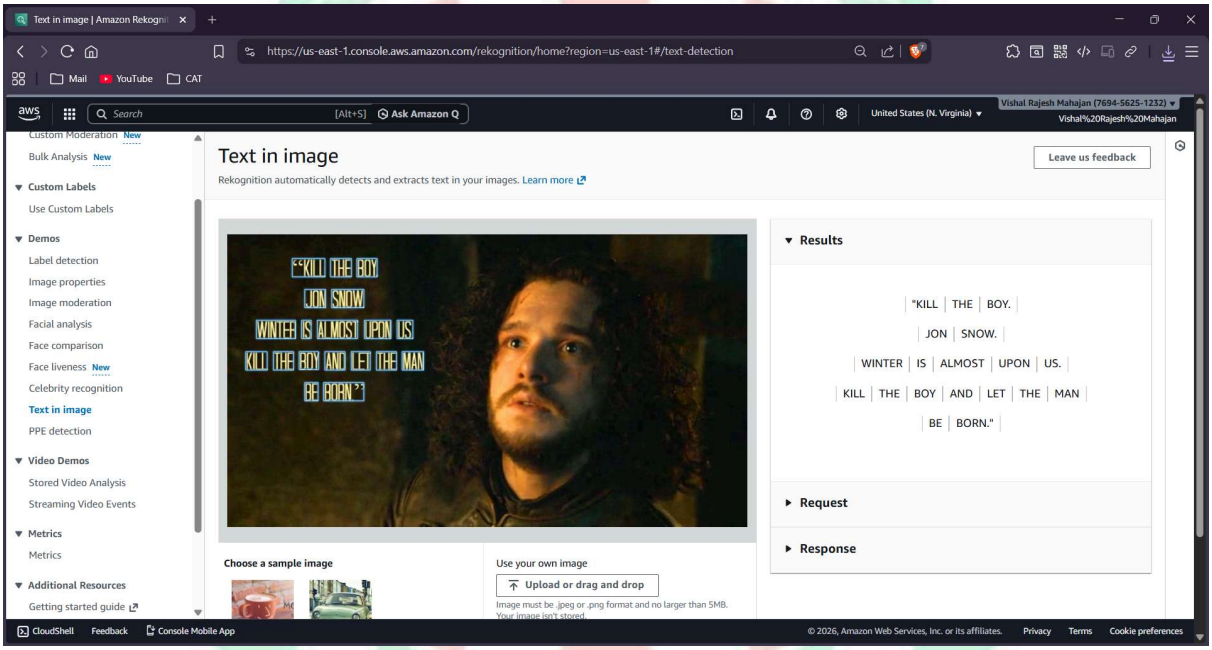
5. Face Liveness



6. Celebrity Recognition



7. Text in Image



8. PPE Detection

